

Batch versus streaming data processing

Previously, you learned about two data processing methods: batch and streaming. **Batch processing** involves collecting large volumes of data over a period of time, then processing it all at once. **Streaming data processing** processes data as it's received. Recognizing the differences between these two processing methods, and how they're used, will help you better understand when and how to apply them. In this reading, you'll learn how frequency informs data processing, and about the differences between processing methods. You'll also learn about common batch and streaming processing tools.

Frequency

Frequency in data processing means how often data is generated or updated by data sources. For example, the frequency of streaming data processing is typically in real-time, or near-real-time. Batch processing might vary based on the specific use case, ranging from hourly, to daily, to weekly, and beyond.

When taking frequency into account, there are several factors that might influence processing, including:

- Business case
- Data requirements
- Data availability
- Latency constraints
- Integration capabilities

Batch vs streaming processing

Batch and streaming processing are both methods for processing data, but are useful for different reasons. By comparing the two, you can better understand when to apply each method:

Processing method	Batch	Streaming
Definition	A method of collecting large volumes of data over a period of time, then processing it all at once	A method of processing data as it's received
Benefits	• Handles large volumes of data efficiently	• Utilizes continuous monitoring to gain timely insights

	• Performs complex analytical calculations • Generates insights for long-term planning and decision making	• Detects anomalies, trends, or critical events in real time • Automates alerts that trigger immediate actions
Challenges	• Data typically requires more storage and more transformation • Insights take longer to generate	• Typically cannot handle large data or deep analysis • Tracking historical trends is more difficult
Example use-cases	• Retail daily sales records • Company payroll	• Medical alert systems • Payment processing

Along with having different use-cases and applications, these processing methods also require different tools. For example, batch processing might use File Transfer Protocol (FTP) or Secure File Transfer Protocols (SFTP) to copy data across cloud buckets in order to move files across systems. FTP is a standard communication protocol that enables the file transfer processes across a computer network. Similarly, SFTP are communication protocols for file transfers often used when processing sensitive data. Batch processing typically takes files as input, then processes them all at once with a programming language or other ETL framework.

Streaming processing uses applications like REST APIs, Pub/Sub, or Kafka to ingest real-time or near real-time data from sources, and processes it to deliver it to the target destination for streaming.

Key takeaways

Batch and streaming data processing are both useful methods for processing data, but they have different uses and requirements. As a cloud data professional, understanding different processing methods will help you identify the best solution for your stakeholders. That way, you can provide them with the data they need to make key decisions.